

PSIR 401-Short Assignment 5

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```
library(readxl)
library(dplyr)
library(tidyr)
library(knitr)
library(kableExtra)
library(ggplot2)
library(scales)

df <- read_excel("/Users/burak/Desktop/assignment5.xlsx")
```

Question 1

Tabulate in turn the anger towards the Supreme Election Council (f1101) [1 “No Anger”, 5 “ALot of Anger toward the SEC”] and probability of voting [0 “Definitely not voting”, 10 “Definitely voting” in the June 23 elections (e03) variables (10 pts).

```
# Anger toward SEC
f1101_labels <- c("1" = "No Anger", "2" = "2", "3" = "3", "4" = "4",
                 "5" = "A Lot of Anger", "99" = "Don't Know/No Answer")

f1101_raw <- as.data.frame(table(Value = df$f1101, useNA = "ifany")) %>%
  rename(Frequency = Freq) %>%
  mutate(
    Label = recode(as.character(Value), !!!f1101_labels, .default =
      ↪ as.character(Value)),
    Percent = round(Frequency / sum(Frequency) * 100, 2),
    `Cum. %` = round(cumsum(Percent), 2)
  ) %>%
  select(Value, Label, Frequency, Percent, `Cum. %`)
```

```
f1101_raw %>%
  kbl(caption = "f1101 - Anger toward the Supreme Election Council (1=No
    ↪ Anger ... 5=A Lot of Anger)", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
    ↪ position = "center", font_size = 11) %>%
  row_spec(0, bold = TRUE)
```

Table 1: f1101 – Anger toward the Supreme Election Council (1=No Anger ... 5=A Lot of Anger)

Value	Label	Frequency	Percent	Cum. %
1	No Anger	365	35.82	35.82
2	2	36	3.53	39.35
3	3	72	7.07	46.42
4	4	131	12.86	59.28
5	A Lot of Anger	390	38.27	97.55
99	Don't Know/No Answer	25	2.45	100.00

```
# Probability of Voting
e03_labels <- c("0" = "Definitely Not Voting", "10" = "Definitely Voting",
  ↪ "99" = "Don't Know/No Answer")

e03_raw <- as.data.frame(table(Value = df$e03, useNA = "ifany")) %>%
  rename(Frequency = Freq) %>%
  mutate(
    Label = recode(as.character(Value), !!!e03_labels, .default =
      ↪ as.character(Value)),
    Percent = round(Frequency / sum(Frequency) * 100, 2),
    `Cum. %` = round(cumsum(Percent), 2)
  ) %>%
  select(Value, Label, Frequency, Percent, `Cum. %`)

e03_raw %>%
  kbl(caption = "e03 - Probability of Voting in June 23 Elections
    ↪ (0=Definitely Not ... 10=Definitely)", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
    ↪ position = "center", font_size = 11) %>%
  row_spec(0, bold = TRUE)
```

Table 2: e03 – Probability of Voting in June 23 Elections (0=Definitely Not ... 10=Definitely)

Value	Label	Frequency	Percent	Cum. %
0	Definitely Not Voting	32	3.14	3.14
1	1	10	0.98	4.12
2	2	5	0.49	4.61
3	3	3	0.29	4.90
4	4	1	0.10	5.00
5	5	31	3.04	8.04
6	6	23	2.26	10.30
7	7	22	2.16	12.46
8	8	57	5.59	18.05
9	9	39	3.83	21.88
10	Definitely Voting	759	74.48	96.36
99	Don't Know/No Answer	37	3.63	99.99

Question 2-3

After recoding missing (i.e., inappropriate response options) data as “system missing” for both variables [99 “Don’t know/No answer”], conduct a chi-squared test (10 pts).

Find out the critical 2 value, and explain whether the association of anger towards the Supreme Election Council decision and the probability of turning out to vote are associated with each other (10 pts).

```
df <- df %>%
  mutate(
    f1101_clean = if_else(f1101 == 99, NA_real_, f1101),
    e03_clean   = if_else(e03 == 99, NA_real_, e03)
  )

cross_tab <- table(`f1101 (Anger->SEC)` = df$f1101_clean, `e03 (Prob.Vote)`
  ↪ = df$e03_clean, useNA = "no")

# ÖNCE: Ki-kare testini çalıştırıp nesneyi oluşturuyoruz
chi_result <- chisq.test(cross_tab)

# SONRA: Oluşan nesnenin içinden parametreleri çekiyoruz
df_chi <- chi_result$parameter
alpha <- 0.05
chi_crit <- qchisq(1 - alpha, df = df_chi)
```

```
# Tablo oluřturma kısmı aynen kalıyor
data.frame(
  Statistic = c("Observed Chi-Squared", "Degrees of Freedom", "Critical
    ↪ Chi-Squared (alpha = 0.05)", "p-value"),
  Value      = c(round(chi_result$statistic, 3), df_chi, round(chi_crit, 3),
    ↪ format.pval(chi_result$p.value, digits = 3))
) %>%
kbl(caption = "Chi-squared Test: Anger toward SEC and Probability of
  ↪ Voting", booktabs = TRUE) %>%
kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
  ↪ position = "center", font_size = 11) %>%
row_spec(0, bold = TRUE)
```

Table 3: Chi-squared Test: Anger toward SEC and Probability of Voting

Statistic	Value
Observed Chi-Squared	83.367
Degrees of Freedom	40
Critical Chi-Squared (alpha = 0.05)	55.758
p-value	6.95e-05

Interpretation: Our calculated Chi-squared value is **83.37**, which is much bigger than the critical value of **55.76**. Also, the p-value is very close to zero ($p < 0.001$).

This means we reject the null hypothesis. There is a **real connection** between anger toward the SEC and the chance of voting. Simply put, people who are angrier about the situation are much more motivated to go and vote.

Question 4

Cross-tabulate the parties that the respondents find themselves closest to (c01) and the probability to vote for Ekrem İmamoğlu (e0402) variables [0 “Definitely would not vote”, 10 “Definitely would vote”] variables (10 pts).

```
party_labels <- c("1" = "AK Parti", "2" = "CHP", "3" = "HDP", "4" = "İYİ
  ↪ Party",
                 "5" = "MHP", "6" = "SP", "90" = "Other", "96" = "None / No
  ↪ Party", "99" = "DK / NA")

df <- df %>%
  mutate(c01_label = recode(as.character(c01), !!!party_labels, .default =
    ↪ "Other"))
```

```

cross_tab_q4 <- table(`c01 (Party)` = df$c01_label, `e0402 (Prob->İmamoğlu)`
  ↪ = df$e0402, useNA = "ifany")

cross_tab_q4 %>%
  as.data.frame.matrix() %>%
  kbl(caption = "Party Closest to Self by Probability to Vote for Imamoglu
  ↪ (Raw)", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
  ↪ position = "center", font_size = 9) %>%
  row_spec(0, bold = TRUE)

```

Table 4: Party Closest to Self by Probability to Vote for Imamoglu (Raw)

	0	1	2	3	4	5	6	7	8	9	10	99
AK Parti	288	55	12	7	5	4	4	1	2	0	10	7
CHP	2	0	0	1	1	5	1	1	11	7	357	2
DK / NA	1	0	3	0	0	3	0	1	3	0	4	2
HDP	6	0	0	0	0	3	0	1	0	0	60	1
IYI Party	0	0	0	0	0	0	1	0	1	1	11	2
MHP	23	4	4	0	0	0	0	0	1	0	3	0
None / No Party	12	1	6	2	1	11	5	0	5	0	23	18
Other	1	0	0	0	0	0	0	0	0	0	1	2
SP	5	1	0	0	0	1	0	0	0	1	0	1

Question 5

Generate a 3-category (trichotomous) variable entitled “pid” (Party ID), where those who find the Justice and Development Party (AK Parti) or the Nationalist Action Party (MHP) closest to themselves score 1, those who find the Republican People’s Party (CHP), the Good Party (IYI), or the Felicity Party (SP) closest score 2, and those who find the People’s Democratic Party (HDP) closest score 3, while all other response categories are coded as missing (10 pts).

```

df <- df %>%
  mutate(
    pid = case_when(
      c01 %in% c(1, 5) ~ 1,      # AK Parti or MHP
      c01 %in% c(2, 4, 6) ~ 2,  # CHP, SP, IYI
      c01 == 3 ~ 3,             # HDP
      TRUE ~ NA_real_
    ),
    pid_label = case_when(

```

```

    pid == 1 ~ "AKP / MHP",
    pid == 2 ~ "CHP / IYI / SP",
    pid == 3 ~ "HDP",
    TRUE     ~ NA_character_
  )
)

pid_tab <- as.data.frame(table(pid = df$pid_label, useNA = "ifany")) %>%
  rename(Frequency = Freq)

pid_tab %>%
  kbl(caption = "Trichotomous Party ID Variable (pid)", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
    ↪ position = "center", font_size = 11) %>%
  row_spec(0, bold = TRUE)

```

Table 5: Trichotomous Party ID Variable (pid)

pid	Frequency
AKP / MHP	430
CHP / IYI / SP	413
HDP	71
NA	105

Question 6

After recoding inappropriate/invalid probability to vote for Ekrem İmamoğlu data (e0402) as (system) missing, report its mean, standard deviation, and frequency (N) for all three groups you coded in the previous step.

```

df <- df %>%
  mutate(e0402_clean = if_else(e0402 == 99, NA_real_, e0402))

desc_q6 <- df %>%
  filter(!is.na(pid)) %>%
  group_by(`Party ID` = pid_label) %>%
  summarise(
    Mean = round(mean(e0402_clean, na.rm = TRUE), 3),
    SD   = round(sd(e0402_clean, na.rm = TRUE), 3),
    N    = sum(!is.na(e0402_clean)),
    .groups = "drop"
  )

```

```

)

desc_q6 %>%
  kbl(caption = "Probability to Vote for Imamoglu by Party ID", booktabs =
    ↪ TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
    ↪ position = "center", font_size = 11) %>%
  row_spec(0, bold = TRUE)

```

Table 6: Probability to Vote for Imamoglu by Party ID

Party ID	Mean	SD	N
AKP / MHP	0.797	2.055	423
CHP / IYI / SP	9.593	1.603	408
HDP	8.886	2.942	70

Question 7

Generate a binary variable (either from the pid or c01 variables), where the HDP supporters score 1, and the People's Alliance (i.e., AK Parti and MHP) supporters score 0. Assuming equal variances, explain whether the difference in their mean probabilities to vote for Ekrem Imamoglu is significant and what the t-test produces differently from the descriptive statistics you presented in response to the previous question.

```

df <- df %>%
  mutate(
    hdp_vs_pa = case_when(
      pid == 3 ~ 1,    # HDP
      pid == 1 ~ 0,    # People's Alliance
      TRUE      ~ NA_real_
    )
  )

q7_data <- df %>% filter(!is.na(hdp_vs_pa) & !is.na(e0402_clean))
ttest_eq <- t.test(e0402_clean ~ hdp_vs_pa, data = q7_data, var.equal = TRUE)

# T-test in kable
data.frame(
  Statistic = c("Mean (People's Alliance = 0)", "Mean (HDP = 1)", "Mean
    ↪ difference", "t-statistic", "Degrees of freedom", "p-value", "95% CI
    ↪ (lower)", "95% CI (upper)"),

```

```

Value      = c(round(ttest_eq$estimate[1], 3), round(ttest_eq$estimate[2],
↪ 3), round(ttest_eq$estimate[1] - ttest_eq$estimate[2], 3),
↪ round(ttest_eq$statistic, 3), round(ttest_eq$parameter, 2),
↪ format.pval(ttest_eq$p.value, digits = 3), round(ttest_eq$conf.int[1],
↪ 3), round(ttest_eq$conf.int[2], 3))
) %>%
kbl(caption = "Equal Variances t-test: Probability to Vote for Imamoglu",
↪ booktabs = TRUE) %>%
kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
↪ position = "center", font_size = 11) %>%
row_spec(0, bold = TRUE)

```

Table 7: Equal Variances t-test: Probability to Vote for Imamoglu

Statistic	Value
Mean (People's Alliance = 0)	0.797
Mean (HDP = 1)	8.886
Mean difference	-8.089
t-statistic	-28.476
Degrees of freedom	491
p-value	<2e-16
95% CI (lower)	-8.647
95% CI (upper)	-7.531

Interpretation:

Simple averages only show the basic numbers, but they cannot prove if a difference is real or just a coincidence. The t-test checks if the gap between the two groups is actually important by looking at the sample size and variation.

Here, the average look at voting for İmamoglu is **0.80** for People's Alliance fans and **8.89** for HDP fans. This **8.09-point gap** is highly significant ($t = -28.48, p < 0.001$). The test proves this big difference is a real fact, not random luck.

Additional visualization

```

q7_data %>%
  mutate(`Party group` = factor(
    hdp_vs_pa,
    levels = c(0, 1),
    labels = c("People's Alliance", "HDP")
  )

```

```

)) %>%
ggplot(aes(x = `Party group`, y = e0402_clean, fill = `Party group`)) +
geom_boxplot(alpha = 0.75, na.rm = TRUE) +
labs(
  x = NULL,
  y = "Probability of voting for İmamoğlu (0-10)"
) +
scale_y_continuous(breaks = 0:10, limits = c(0, 10)) +
theme_minimal() +
theme(legend.position = "none")

```

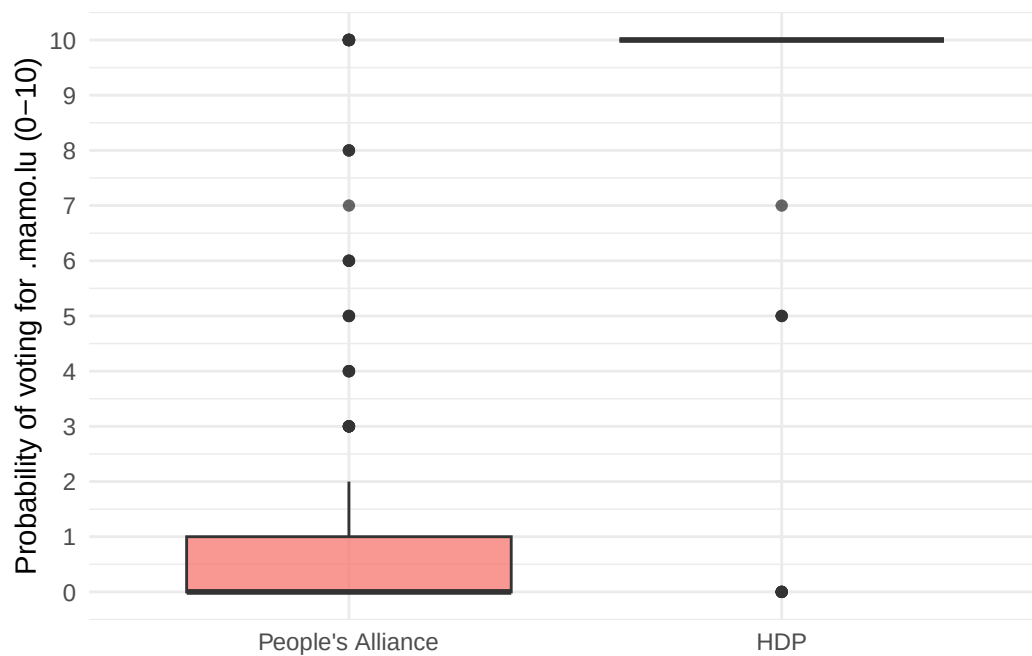


Figure 1: Probability of voting for İmamoğlu by party group

Question 8

Conduct a t-test assuming unequal variances, and briefly explain what the mean probabilities for those two groups of respondents and their difference suggest (10 pts).

```

ttest_uneq <- t.test(e0402_clean ~ hdp_vs_pa, data = q7_data, var.equal =
  ↪ FALSE)

```

```

data.frame(
  Statistic = c("Mean (People's Alliance = 0)", "Mean (HDP = 1)", "Mean
    ↪ difference", "t-statistic", "Degrees of freedom", "p-value", "95% CI
    ↪ (lower)", "95% CI (upper)"),
  Value      = c(round(ttest_uneq$estimate[1], 3),
    ↪ round(ttest_uneq$estimate[2], 3), round(ttest_uneq$estimate[1] -
    ↪ ttest_uneq$estimate[2], 3), round(ttest_uneq$statistic, 3),
    ↪ round(ttest_uneq$parameter, 2), format.pval(ttest_uneq$p.value, digits
    ↪ = 3), round(ttest_uneq$conf.int[1], 3), round(ttest_uneq$conf.int[2],
    ↪ 3))
) %>%
kbl(caption = "Unequal Variances t-test", booktabs = TRUE) %>%
kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
  ↪ position = "center", font_size = 11) %>%
row_spec(0, bold = TRUE)

```

Table 8: Unequal Variances t-test

Statistic	Value
Mean (People's Alliance = 0)	0.797
Mean (HDP = 1)	8.886
Mean difference	-8.089
t-statistic	-22.13
Degrees of freedom	80.51
p-value	<2e-16
95% CI (lower)	-8.816
95% CI (upper)	-7.362

Interpretation: The Welch t-test gives the exact same result even without assuming equal variations ($t = -22.13, p < 0.001$).

This shows clear **strategic voting**. Even though HDP voters do not share the same political ideas with the CHP, they almost completely voted for İmamoğlu to stand against the AKP-MHP alliance.

Question 9

After recoding the “Don’t Know/No Answer” category as missing for the probability to vote for Binali Yıldırım variable (e0401), generate a dichotomous variable entitled “akp dum” from the party identification (c05) variable, where those who reported to be an AKParti partisan score 1 and those who reported to be an MHP partisan score 0 (10 pts). Then, calculate the

mean probabilities of voting for Binali Yıldırım of AK Parti and MHP supporters with varying levels of anger toward the Supreme Election Council (10 pts).

```
df <- df %>%
  mutate(
    e0401_clean = if_else(e0401 == 99, NA_real_, e0401),
    akp_dum = case_when(
      c05 == 1 ~ 1,    # AK Parti partisan
      c05 == 5 ~ 0,    # MHP partisan
      TRUE      ~ NA_real_
    )
  )

# (Wide Pivot)
means_q9 <- df %>%
  filter(!is.na(akp_dum) & !is.na(f1101_clean) & !is.na(e0401_clean)) %>%
  group_by(Party = if_else(akp_dum == 1, "Mean (AKP)", "Mean (MHP)"),
    ↪ f1101_clean) %>%
  summarise(Mean_e0401 = round(mean(e0401_clean, na.rm = TRUE), 3), .groups =
    ↪ "drop") %>%
  tidyr::pivot_wider(names_from = f1101_clean, values_from = Mean_e0401) %>%
  rename(`Anger toward SEC (1 No, 5 A Lot)` = Party)

means_q9 %>%
  kbl(caption = "Mean Probability to Vote for Yildirim by Anger toward SEC
    ↪ and Party", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
    ↪ position = "center", font_size = 11) %>%
  row_spec(0, bold = TRUE)
```

Table 9: Mean Probability to Vote for Yildirim by Anger toward SEC and Party

Anger toward SEC (1 No, 5 A Lot)	1	2	3	4	5
Mean (AKP)	9.727	8.739	8.955	8.621	7.211
Mean (MHP)	9.158	8.500	6.200	5.250	4.400

Question 10

Plot the mean probabilities (as in Figure 7 in the “...Sample Excel File” document and in the previous assignment). Briefly explain what the mean probabilities you calculated and plotted suggest (10 pts).

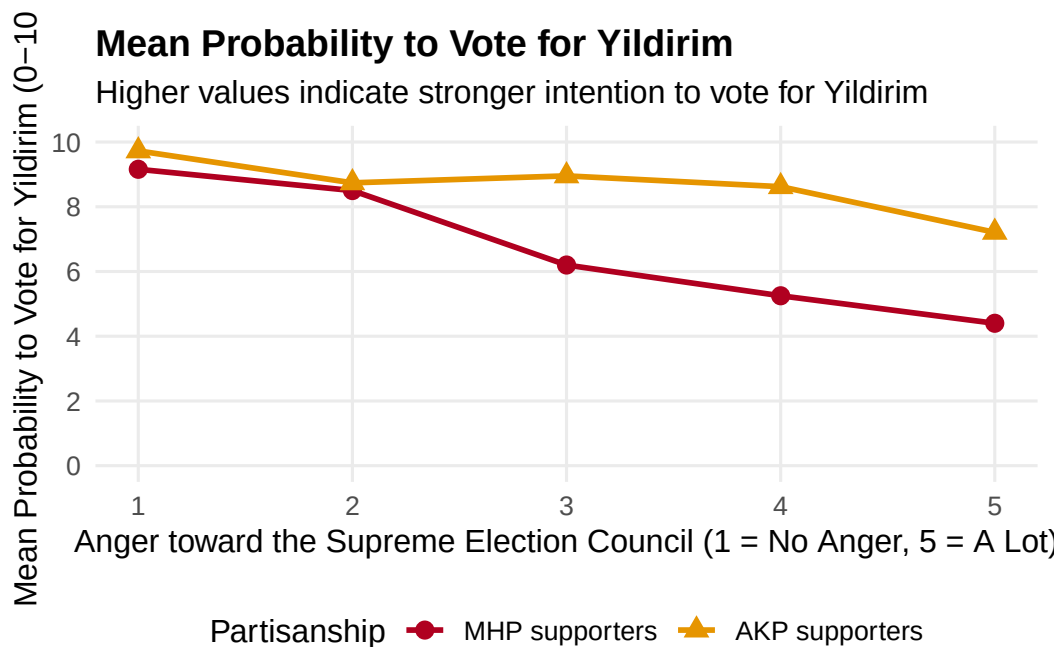
```

plot_data <- df %>%
  filter(!is.na(akp_dum) & !is.na(f1101_clean) & !is.na(e0401_clean)) %>%
  group_by(f1101_clean, akp_dum) %>%
  summarise(Mean_e0401 = mean(e0401_clean), .groups = "drop") %>%
  mutate(Partisanship = factor(akp_dum, levels = c(0, 1), labels = c("MHP
    ↳ supporters", "AKP supporters")))

p10 <- ggplot(plot_data, aes(x = f1101_clean, y = Mean_e0401, color =
  ↳ Partisanship, group = Partisanship, shape = Partisanship)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  scale_x_continuous(name = "Anger toward the Supreme Election Council (1 =
    ↳ No Anger, 5 = A Lot)", breaks = 1:5) +
  scale_y_continuous(name = "Mean Probability to Vote for Yildirim (0-10)",
    ↳ limits = c(0, 10), breaks = seq(0, 10, 2)) +
  scale_color_manual(values = c("AKP supporters" = "#E69500", "MHP
    ↳ supporters" = "#B00020")) + # Senin Orjinal Renklerin
  labs(title = "Mean Probability to Vote for Yildirim", subtitle = "Higher
    ↳ values indicate stronger intention to vote for Yildirim") +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom", plot.title = element_text(face = "bold"),
    ↳ panel.grid.minor = element_blank())

print(p10)

```



```
ggsave("Q10_mean_prob_plot.png", plot = p10, width = 7, height = 5, dpi =
  ↪ 150)
```

Interpretation: The line graph shows how anger changes the vote for Binali Yildirim:

- **AK Parti Voters:** They stay very loyal to Yildirim (scoring between 7 and 10) no matter how angry they are about the SEC. Their party choice did not change at all.
- **MHP Voters:** Their support drops quickly. When they have “No Anger” the score is 9.16, but with “A Lot of Anger” it drops to 4.40.

Note: The MHP group is very small ($n = 35$) , so this drop might not be 100% reliable.

Bonus Question

Plot histograms (in percentages) of anger toward the Supreme Election Council for both the AK Parti and MHP supporters, and briefly explain what the two plots show (Bonus question–10 pts).

```
bonus_tab <- df %>%
  filter(!is.na(akp_dum) & !is.na(f1101_clean)) %>%
  mutate(Group = factor(akp_dum, levels = c(1, 0), labels = c("AKP
  ↪ supporters", "MHP supporters"))) %>%
```

```

group_by(Group, f1101_clean) %>%
summarise(n = n(), .groups = "drop_last") %>%
mutate(pct = n / sum(n) * 100) %>%
ungroup()

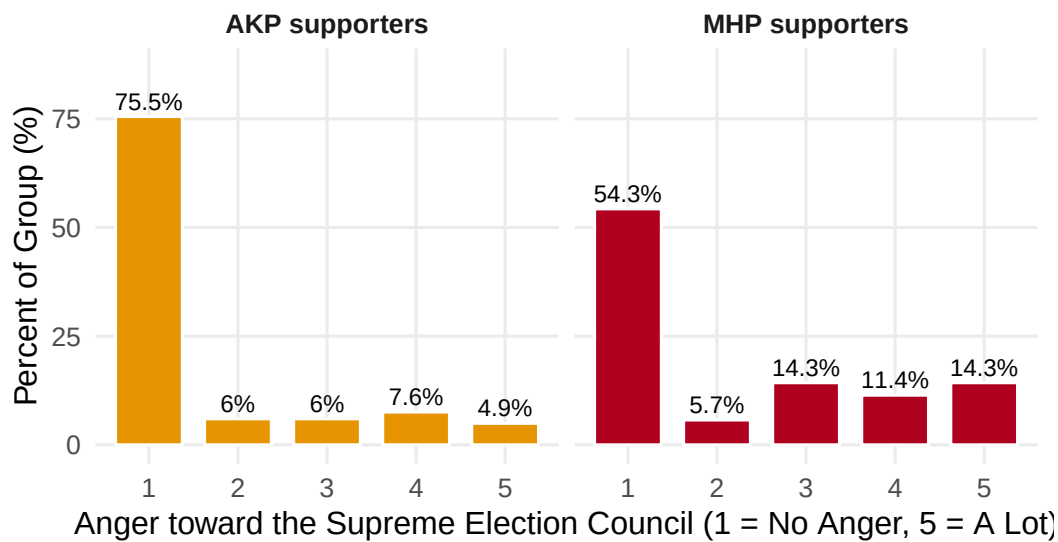
p_bonus <- ggplot(bonus_tab, aes(x = factor(f1101_clean), y = pct, fill =
  ↳ Group)) +
  geom_col(show.legend = FALSE, color = "white", width = 0.75) +
  facet_wrap(~ Group) +
  scale_fill_manual(values = c("AKP supporters" = "#E69500", "MHP supporters"
    ↳ = "#B00020")) + # Senin Orijinal Renklerin
  geom_text(aes(label = paste0(round(pct, 1), "%")), vjust = -0.4, size = 3)
  ↳ +
  scale_y_continuous(limits = c(0, max(bonus_tab$pct) * 1.15)) +
  labs(x = "Anger toward the Supreme Election Council (1 = No Anger, 5 = A
    ↳ Lot)", y = "Percent of Group (%)",
        title = "Distribution of Anger toward the Supreme Election Council",
        ↳ subtitle = "Within-group percentages, by bypassing") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"), panel.grid.minor =
    ↳ element_blank(), strip.text = element_text(face = "bold"))

print(p_bonus)

```

Distribution of Anger toward the Supreme Election Co

Within-group percentages, by bypassing



Interpretation: The charts show that AKP and MHP voters felt completely different about the SEC decision:

- **AK Parti Group:** Most of them (**75.5%**) felt zero anger (Category 1). They completely agreed with the decision.
- **MHP Group:** Only **54.3%** felt zero anger. The rest are spread out, and many felt medium or high anger.

This means the MHP base had mixed feelings about the decision, while the AKP base was totally happy with it.